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Ethnomedicinal plants used against tuberculosis by the ethnic people of Ranchi district of Jharkhand

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Abstract : Tuberculosis is a wide spread disease in Ranchi district, it is a common infectious disease but in many cases is very lethal. There are several plants used by the tribal peoples to cure this disease .Ethnomedicine deals with the nature and application of plants used as traditional medicine system and it involves the identification of the plants used in the traditional remedies. Often Ethnobotanical inventories are made by conducting interviews with the tribal people. Formal surveys in the field and by collection and identification of plants specimen used. This article deals with the several plant species used in treatment of tuberculosis by ethnic people of Ranchi district and 30 plants and their way of administration has been described in the present paper.

Keywords : Ethno medicine, Tuberculosis, Tribes, *Mycobacterium tuberculosis*

INTRODUCTION

Ranchi the capital of Jharkhand which is in Ranchi district is divided into 18 blocks and 1311 villages according to census 2011[1]. Most of the population lives in villages. The main tribes concentrated in these areas are Oraon, Munda, Ho, Kharwar, Lohra, Bhumij, kharia and others.

Tuberculosis (TB) is an infectious disease that is caused by a bacterium called *Mycobacterium tuberculosis*[2]. The drug currently used to treat TB are mainly RIFAMPICIN, EHAMBUTOL, ISONIAZID and PYRAZINAMIDE but the emergence of multiple drug resistant (MDR) strains of *M. tuberculosis* (defined resistance against isoneazid and rifampin)is now common in number of patients because of uncontrolled application of these anti-tuberculosis drugs[3].

The medicinal properties of plants have been well known from ancient time and plant offers a source of potent anti-microbial agents in the form of secondary metabolites[4]. Anti tuberculosis activities have been reported in the number of plants[5]. According to Ayurveda tuberculosis is known as RAJAYAKHSMA, YAKSHMA, SHOOSHA, and KSHAYA[6]. The present paper contributes to Ethnomedicinal botany regarding treatment of the said disease tuberculosis.

MATERIALS & METHODS

Anti- tuberculosis allopathic drugs have been prescribed to controlled symptoms of this disease but they result into side effects like hepatitis, hypersensitivity reactions, nausea, vomiting etc [7, 8, 9], thus the herbal medicines are used for tuberculosis.Ehnobotanical surveys were conducted for the information of abundance and therapeutical properties of the wild plants for the treatment of tuberculosis [10, 15, 16]. Different plant species were collected after making a critical observation and conversation with the tribes. Firstly field trips were conducted and local vaidyas were contacted and also the people suffering from TB and herbal practitioners were questioned. Questionnaire conducted for first hand

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information regarding the medicinal plants was gathered. Detailed information regarding botanical identification of plants[12], local names, part used for the drug[11 , 13,14] was recorded, herbaria was deposited with all relevant informations in "Department of Botany, Ranchi College Ranchi".

Table 1: List of anti -tuberculosis plants used by ethnic population of ranchi district and their way of administration:

Sl. No.	BOTANICAL NAME	FAMILY	COMMON NAME	PLANT PART USED/ METHOD OF USING
1.	*Adhatoda vasica L.*	Acanthaceae	Vasaka/basaka	Oil obtained from leaf, flowers and root. Active principal vasicine
2.	*Allium cepa L.*	Liliaceae	Onion/pyaz	Fresh juice of bulb
3.	*Aloe vera L.*	Liliaceae	Aloevera /ghigvar	Extract of leaf
4.	*Allium sativum L.*	Liliaceae	Garlic /lahsun	Cloves
6.	*Acacia catechu (L.)willd*	Mimosaceae	Kala khair/kattha	Fresh root pest adviced to take orally with empty stomach for 60 days
6.	*Ailanthus exelsa Roxb.*	Simaroubaceae	Ailanthus	Root
7.	*Aegle marmelos Corr.*	Rutaceae	Bel	Leaf extract is prescribed
8.	*Andrographis paniculata Nees.*	Acanthaceae	Kalmegha	Shade dried powder is adviced to take for preparation of decoction with two glass water reduced till the half then taken orally twise in a day.
9.	*Datura metel L.*	Solanaceae	Kala dhatura	Leaf extract is prescribed.
10.	*Ocimum sanctum L.*	Labiateae	Holy basil/Tulsi	Leaf extract.
11.	*Piper nigrum L.*	Piperaceae	Black pepper/ kali mirch	Fruit
12.	*Acorus calamus L.*	Araceae	Sweet flag/ vekhand	Rhizome
13.	*Pueraria tuberosa DC.*	Fabaceae	Patal kohra/ bidaarikand	Tubers
14.	*Moringa olifera Lam.*	Moringaceae	Drumstick/sahjan	Seeds
15.	*Tamarindus indica L.*	Fabaceae	Tamarind/imli	Fruit
16.	*Acacia nilotica L.*	Mimoseae	Kikar/babul	Root
17.	*Adonsonia digitata L*	Malvaceae	Kalptath	Seed
18.	*Eucalyptus globues Labill.*	Myrtaceae	Eucalyptus	Essential oil
19.	*Carica papaya L.*	Caricaceae	Papita	Leaves
20.	*Cedrus deodara Lamb.*	Pinaceae	Deodar	Oil
21.	*Ocimum sanctum L.*	Labiateae	Tulsi	Leaves decoction
22.	*Centella asiatica (L.) Urb.*	Apiaceae	Beng saag	Leaves
23.	*Catheranthus roseus (L.) G. Don*	Apocynaceae	Sada bahar	Leaves
24.	*Holarrhena pubescens Wall ex G.Don*	Apocynaceae	Kurchi	Root, bark and seeds
25.	*Colocasia esculenta (L.) Schott*	Araceae	Taro	Corm
26.	*Calendula officinalis L.*	Asteraceae	Marigold/ Gendaful	Leaf, oil
27.	*Ricinus communis L.*	Euphorbiaceae	Castor/arundy	Seed
28.	*Swertia chirayita Roxb.*	Gentianaceae	Chirayata	Whole plant
29.	*Swietenia mahogany (L.) Jacq.*	Meliaceae	Mahogany	Dry bark powder
30.	*Tinospora cordifolia (Thunb.) Miers*	Menispermaceae	Amrita /guduchi	Stem and leaf

RESULTS & DISCUSSION

The survey of different blocks and villages of Ranchi district of Jharkhand revealed that the local population are using many local wild plant species for the treatment of tuberculosis . After the survey we found that most of the herbal medicines used by the tribals are obtained from the near by forest and are local wild weeds. They are taken mostly in the form of juices, pest, dry powder or ashes.

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